Name: _____ **Date:** _____

Software Engineering Specialization

Exercise: Modeling Software Systems using UML

We want to represent information about a graphical editor that supports grouping. A document is composed of several sheets. Each sheet can contain several drawing objects, which can be text, geometric objects or groups. A group is simply a collection of drawing objects, possibly including other groups. A group contains two or more drawing objects. A drawing object can be a member of at most one group. Each sheet belongs to at most one document and each drawing object belongs to exactly one sheet.

In the space below, construct a class diagram for the graphical editor, that shows, as necessary, associations, aggregations, compositions, generalizations among the classes as well as any necessary association classes and constraints. Associations should be named, if necessary. Using the problem statement and real-world knowledge, give the most likely multiplicities for each association, aggregation and composition. If a multiplicity cannot be inferred from the problem statement or real-world domain knowledge, then indicate this with a "?".